Proyecto Final Redes I

Alumno: Iván Augusto Camargo López Código 2230033

Eres el ingeniero de redes encargado de diseñar y configurar la infraestructura de una empresa tecnológica. Tu objetivo es crear una red compuesta por 6 subredes totalmente conectadas y funcionales, aplicando los conocimientos de VLSM, enrutamiento estático, DHCP, DNS, HTTP, FTP y tolerancia a fallos.

Requerimientos Técnicos

1. Seleccionar una red base y calcular las 6 subredes mediante VLSM.
2. Configurar 5 routers 2911 en una red mayada.
3. Cada red debe tener al menos 3 dispositivos (PC, laptop, IP Phone, impresora, etc.).
4. Una de las redes tendrá los servidores DNS, HTTP y FTP. Con alojamiento de la página principal de la empresa y esta debe accederse desde cualquier PC
5. Una red será únicamente WiFi con DHCP habilitado.
6. Cada red (excepto la WiFi) debe tener un router inalámbrico adicional.
7. Todas las redes deben poder comunicarse entre sí.
8. Crear un usuario y password para FTP el cual podrá accedre desde cualquier equipo de cualquier red.
9. Todos los routers deben guardar su configuración startup-config en el servidor FTP.
10. Debe existir resiliencia ante la caída de enlaces.
11. Entregar el archivo Packet Tracer (.pkt) y este documento completado con evidencias.

1. Tabla de Subredes (VLSM)

Subred	Dirección de red	Gateway	Broadcast	Máscara	Hosts válidos
LAN1	9.9.9.0	9.9.9.1	9.9.9.31	/27 255.255.255.224	9.9.9.1 – 9.9.9.30
LAN2	9.9.9.32	9.9.9.33	9.9.9.63	/27 255.255.255.224	9.9.9.33 – 9.9.9.62
LAN3	9.9.9.64	9.9.9.65	9.9.9.95	/27 255.255.255.224	9.9.9.65 – 9.9.9.94
LAN4	9.9.9.96	9.9.9.97	9.9.9.127	/27 255.255.255.224	9.9.9.97 – 9.9.9.126
LAN5	9.9.9.128	9.9.9.129	9.9.9.159	/27 255.255.255.224	9.9.9.129 – 9.9.9.158
LAN6	9.9.9.160	9.9.9.161	9.9.9.191	/27 255.255.255.224	9.9.9.161 – 9.9.9.190
ROUTERS_1	9.9.9.192	9.9.9.193 9.9.9.194	9.9.9.195	/30 255.255.255.252	9.9.9.193 - 9.9.9.194
ROUTERS_2	9.9.9.196	9.9.9.197 9.9.9.198	9.9.9.199	/30 255.255.255.252	9.9.9.197 - 9.9.9.198
ROUTERS_3	9.9.9.200	9.9.9.201 9.9.9.202	9.9.9.203	/30 255.255.255.252	9.9.9.201 - 9.9.9.202

ROUTERS_4	9.9.9.204	9.9.9.205 9.9.9.206	9.9.9.207	/30 255.255.255.252	9.9.9.205 - 9.9.9.206
ROUTERS_5	9.9.9.208	9.9.9.209 9.9.9.210	9.9.9.211	/30 255.255.255.252	9.9.9.209 - 9.9.9.210
ROUTERS_6	9.9.9.212	9.9.9.213 9.9.9.214	9.9.9.215	/30 255.255.255.252	9.9.9.213 - 9.9.9.214
ROUTERS_7	9.9.9.216	9.9.9.217 9.9.9.218	9.9.9.219	/30 255.255.255.252	9.9.9.217 - 9.9.9.218
ROUTERS_8	9.9.9.220	9.9.9.221 9.9.9.222	9.9.9.223	/30 255.255.255.252	9.9.9.221 - 9.9.9.222
ROUTERS_9	9.9.9.224	9.9.9.225 9.9.9.226	9.9.9.227	/30 255.255.255.252	9.9.9.225 - 9.9.9.226
ROUTERS_10	9.9.9.228	9.9.9.229 9.9.9.230	9.9.9.231	/30 255.255.255.252	9.9.9.229 - 9.9.9.230

2. Tabla de Subredes (se particionaron en subredes de respaldo – no usadas pero podrían servir de apoyo a la empresa en algún momento)

Subred	Dirección de red	Gateway	Broadcast	Máscara	Hosts válidos
LAN_1_1	9.9.9.0	9.9.9.1	9.9.9.15	/28 255.255.255.240	9.9.9.1 – 9.9.9.14
LAN_1_2	9.9.9.16	9.9.9.17	9.9.9.31	/28 255.255.255.240	9.9.9.17 – 9.9.9.30
LAN_2_1	9.9.9.32	9.9.9.33	9.9.9.47	/28 255.255.255.240	9.9.9.33 – 9.9.9.46
LAN_2_2	9.9.9.48	9.9.9.49	9.9.9.63	/28 255.255.255.240	9.9.9.49 – 9.9.9.62
LAN_3_1	9.9.9.64	9.9.9.65	9.9.9.79	/28 255.255.255.240	9.9.9.65 – 9.9.9.78
LAN_3_2	9.9.9.80	9.9.9.81	9.9.9.95	/28 255.255.255.240	9.9.9.81 – 9.9.9.94
LAN_4_1	9.9.9.96	9.9.9.97	9.9.9.111	/28 255.255.255.240	9.9.9.97 – 9.9.9.110
LAN_4_2	9.9.9.112	9.9.9.113	9.9.9.127	/28 255.255.255.240	9.9.9.113 – 9.9.9.126
LAN_5_1	9.9.9.128	9.9.9.129	9.9.9.143	/28 255.255.255.240	9.9.9.129 – 9.9.9.142
LAN_5_2	9.9.9.144	9.9.9.145	9.9.9.159	/28 255.255.255.240	9.9.9.145 – 9.9.9.158
LAN6	9.9.9.160	9.9.9.161	9.9.9.191	/27 255.255.255.224	9.9.9.161 – 9.9.9.190
ROUTERS_1	9.9.9.192	9.9.9.193 9.9.9.194	9.9.9.195	/30 255.255.255.252	9.9.9.193 - 9.9.9.194
ROUTERS_2	9.9.9.196	9.9.9.197 9.9.9.198	9.9.9.199	/30 255.255.255.252	9.9.9.197 - 9.9.9.198
ROUTERS_3	9.9.9.200	9.9.9.201 9.9.9.202	9.9.9.203	/30 255.255.255.252	9.9.9.201 - 9.9.9.202
ROUTERS_4	9.9.9.204	9.9.9.205 9.9.9.206	9.9.9.207	/30 255.255.255.252	9.9.9.205 - 9.9.9.206
ROUTERS_5	9.9.9.208	9.9.9.209 9.9.9.210	9.9.9.211	/30 255.255.255.252	9.9.9.209 - 9.9.9.210
ROUTERS_6	9.9.9.212	9.9.9.213 9.9.9.214	9.9.9.215	/30 255.255.255.252	9.9.9.213 - 9.9.9.214
ROUTERS_7	9.9.9.216	9.9.9.217 9.9.9.218	9.9.9.219	/30 255.255.255.252	9.9.9.217 - 9.9.9.218
ROUTERS_8	9.9.9.220	9.9.9.221 9.9.9.222	9.9.9.223	/30 255.255.255.252	9.9.9.221 - 9.9.9.222
ROUTERS_9	9.9.9.224	9.9.9.225 9.9.9.226	9.9.9.227	/30 255.255.255.252	9.9.9.225 - 9.9.9.226
ROUTERS_10	9.9.9.228	9.9.9.229 9.9.9.230	9.9.9.231	/30 255.255.255.252	9.9.9.229 - 9.9.9.230

3. Dispositivos por Red

Red	Dispositivo	Tipo	IP asignada	Observaciones	Interfaz Router
LAN1_1	PC0	Cableado	9.9.9.2		g0/0
LAN1_1	Laptop0	Cableado	9.9.9.3		g0/0
LAN1_1	PC1	Cableado	9.9.9.4		g0/0
LAN1_1	Wireless Router0	Cableado	-	Access Point	g0/0
LAN1_2	Wireless Router1	Cableado	-	Respaldo Wireless	
LAN 2_1	Printer3	Cableado	9.9.9.34		g0/1
LAN 2_1	Laptop2	Cableado	9.9.9.35		g0/1
LAN 2_1	PC4	Cableado	9.9.9.36		g0/1
LAN 2_1	Wireless Router 2	Cableado	-	Access Point	g0/1
LAN 2_2	Wireless Router 3	Cableado	-	Respaldo Wireless	
LAN 3_1	Printer0	Cableado	9.9.9.66		g0/0
LAN 3_1	Printer1	Cableado	9.9.9.67		g0/0
LAN 3_1	Printer2	Cableado	9.9.9.68		g0/0
LAN 3_1	Wireless Router 5	Cableado	-	Access Point	g0/0
LAN 3_2	Wireless Router 6	Cableado	-	Respaldo Wireless	
LAN 4_1	Laptop1	Cableado	9.9.9.98		g0/0
LAN 4_1	PC2	Cableado	9.9.9.99		g0/0
LAN 4_1	PC3	Cableado	9.9.9.100		g0/0
LAN 4_1	Wireless Router 10	Cableado	-	Access Point	g0/0
LAN 4_2	Wireless Router 7	Cableado	-	Respaldo Wireless	
LAN 5_1	Server_FTP	Cableado	9.9.9.130	Servidor FTP	g0/0
LAN 5_1	Server_HTTP	Cableado	9.9.9.131	Servidor HTTP	g0/0
LAN 5_1	Server_DNS	Cableado	9.9.9.132	Servidor DNS	g0/0
LAN 5_1	Wireless Router 9	Cableado	9.9.9.133	Access Point	g0/0

LAN_5_1	PC5	Cableado	9.9.9.134		g0/0
LAN_5_1	PC6	Cableado	9.9.9.135		g0/0
LAN_5_1	PC7	Cableado	9.9.9.136		g0/0
LAN 5_2	Wireless Router 8	Cableado	-	Respaldo Wireless	
LAN 6	Access Point 1	Cableado	-	Access Point (sin cableado)	g0/0
LAN 6	SmartPhone0	Inalámbrico	9.9.9.162	Rango HDCP 9.9.9.162 – 9.9.9.190	g0/0
LAN 6	SmartPhone1	Inalámbrico	9.9.9.163	Rango HDCP 9.9.9.162 – 9.9.9.190	g0/0
LAN 6	Tablet PC0	Inalámbrico	9.9.9.164	Rango HDCP 9.9.9.162 – 9.9.9.190	g0/0

4. Malla de Routers

Conexión	Router_Bucaramanga	Router_Malaga	Router_Barbosa	Router_Floridablanca	Router_Piedecuesta
Router_Bucaramanga	-	s0/1/0 (Bucaramanga) – s0/1/0 (Malaga)	s0/1/1 (Bucaramanga) – s0/1/0 (Barbosa)	s0/2/0 (Bucaramanga) – g0/1 (Floridablanca)	s0/2/1 (Bucaramanga) – s0/1/0 (Piedecuesta
Router_Malaga	s0/1/0 (Bucaramanga) – s0/1/0 (Malaga)	-	s0/1/1 (Malaga) – s0/1/1 (Barbosa)	s0/2/0 (Malaga) – s0/1/1 (Floridablanca)	s0/2/1 (Malaga) – s0/1/1 (Piedecuesta)
Router_Barbosa	s0/1/1 (Bucaramanga) – s0/1/0 (Barbosa)	s0/1/1 (Malaga) – s0/1/1 (Barbosa)	-	s0/2/0 (Barbosa) – s0/2/0 (/Floridablanca)	s0/2/1 (Barbosa) – s0/2/0 (Piedecuesta)
Router_Floridablanca	s0/2/0 (Bucaramanga) – g0/1 (Floridablanca)	s0/2/0 (Malaga) – s0/1/1 (Floridablanca)	s0/2/0 (Barbosa) – s0/2/0 (/Floridablanca)	-	s0/2/1 (Floridablanca) – s0/2/1 (Piedecuesta)
Router_Piedecuesta	s0/2/1 (Bucaramanga) – s0/1/0 (Piedecuesta	s0/2/1 (Malaga) – s0/1/1 (Piedecuesta)	s0/2/1 (Barbosa) – s0/2/0 (Piedecuesta)	s0/2/1 (Floridablanca) – s0/2/1 (Piedecuesta)	-

5. Tabla de Claves de router

Router	password	secret
Router_Bucaramanga	Pass1	Secret1
Router_Malaga	Pass2	Secret2
Router_Barbosa	Pass3	Secret3
Router_Floridablanca	Pass4	Secret4
Router_Piedecuesta	Pass5	Secret5

6. Configuración de Servidores

• Servidor DNS IP: 9.9.9.132

• Servidor HTTP IP: 9.9.9.131Dirección WEB: www.pruebaredes.com

• Servidor FTP IP: 9.9.9.130

• Usuario FTP: pruebaClase Clave: prueba123

4. Evidencias del Desarrollo

A continuación, incluye los pantallazos de cada fase del proyecto:

- Configuración de routers (show ip interface brief / show ip route)

Router_Bucaramanga

```
Router_Bucaramanga>en
Password:
Router_Bucaramanga#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     9.9.9.1         YES NVRAM  up                    up
GigabitEthernet0/1     9.9.9.33        YES NVRAM  up                    up
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/0   unassigned      YES NVRAM  administratively down down
Serial0/1/0            9.9.9.193       YES NVRAM  up                    up
Serial0/1/1            9.9.9.197       YES NVRAM  up                    up
Serial0/2/0            9.9.9.201       YES NVRAM  up                    up
Serial0/2/1            9.9.9.205       YES NVRAM  up                    up
Serial0/3/0            unassigned      YES NVRAM  down                  down
Serial0/3/1            unassigned      YES NVRAM  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Router_Bucaramanga#
```

```
Router_Bucaramanga#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     9.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
C       9.9.9.0/28 is directly connected, GigabitEthernet0/0
L       9.9.9.1/32 is directly connected, GigabitEthernet0/0
C       9.9.9.32/28 is directly connected, GigabitEthernet0/1
L       9.9.9.33/32 is directly connected, GigabitEthernet0/1
S       9.9.9.64/28 [1/0] via 9.9.9.194
S       9.9.9.96/28 [1/0] via 9.9.9.198
S       9.9.9.128/28 [1/0] via 9.9.9.202
S       9.9.9.160/27 [1/0] via 9.9.9.206
C       9.9.9.192/30 is directly connected, Serial0/1/0
L       9.9.9.193/32 is directly connected, Serial0/1/0
C       9.9.9.196/30 is directly connected, Serial0/1/1
L       9.9.9.197/32 is directly connected, Serial0/1/1
C       9.9.9.200/30 is directly connected, Serial0/2/0
L       9.9.9.201/32 is directly connected, Serial0/2/0
C       9.9.9.204/30 is directly connected, Serial0/2/1
L       9.9.9.205/32 is directly connected, Serial0/2/1

Router_Bucaramanga#
```

Router_Malaga

```
Router_Malaga>en
Password:
Router_Malaga#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     9.9.9.65        YES NVRAM  up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  down                  down
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/0   unassigned      YES unset  down                  down
Serial0/1/0            9.9.9.194       YES manual up                    up
Serial0/1/1            9.9.9.209       YES manual up                    up
Serial0/2/0            9.9.9.213       YES manual up                    up
Serial0/2/1            9.9.9.217       YES manual up                    up
Serial0/3/0            unassigned      YES unset  down                  down
Serial0/3/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Router_Malaga#
```

```
Router_Malaga#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     9.0.0.0/8 is variably subnetted, 15 subnets, 4 masks
S       9.9.9.0/28 [1/0] via 9.9.9.193
S       9.9.9.32/28 [1/0] via 9.9.9.193
C       9.9.9.64/28 is directly connected, GigabitEthernet0/0
L       9.9.9.65/32 is directly connected, GigabitEthernet0/0
S       9.9.9.96/28 [1/0] via 9.9.9.210
S       9.9.9.128/28 [1/0] via 9.9.9.214
S       9.9.9.160/27 [1/0] via 9.9.9.218
C       9.9.9.192/30 is directly connected, Serial0/1/0
L       9.9.9.194/32 is directly connected, Serial0/1/0
C       9.9.9.208/30 is directly connected, Serial0/1/1
L       9.9.9.209/32 is directly connected, Serial0/1/1
C       9.9.9.212/30 is directly connected, Serial0/2/0
L       9.9.9.213/32 is directly connected, Serial0/2/0
C       9.9.9.216/30 is directly connected, Serial0/2/1
L       9.9.9.217/32 is directly connected, Serial0/2/1

Router_Malaga#
Router_Malaga#
```

Router_Barbosa

```
Router_Barbosa#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     9.9.9.97        YES NVRAM  up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/0   unassigned      YES unset  down                  down
Serial0/1/0            9.9.9.198       YES manual up                    up
Serial0/1/1            9.9.9.210       YES manual up                    up
Serial0/2/0            9.9.9.221       YES manual up                    up
Serial0/2/1            9.9.9.225       YES manual up                    up
Serial0/3/0            unassigned      YES unset  down                  down
Serial0/3/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Router_Barbosa#
```

```
Vlan1                  unassigned      YES unset  administratively down down
Router_Barbosa#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     9.0.0.0/8 is variably subnetted, 15 subnets, 4 masks
S       9.9.9.0/28 [1/0] via 9.9.9.197
S       9.9.9.32/28 [1/0] via 9.9.9.197
S       9.9.9.64/28 [1/0] via 9.9.9.209
C       9.9.9.96/28 is directly connected, GigabitEthernet0/0
L       9.9.9.97/32 is directly connected, GigabitEthernet0/0
S       9.9.9.128/28 [1/0] via 9.9.9.222
S       9.9.9.160/27 [1/0] via 9.9.9.226
C       9.9.9.196/30 is directly connected, Serial0/1/0
L       9.9.9.198/32 is directly connected, Serial0/1/0
C       9.9.9.208/30 is directly connected, Serial0/1/1
L       9.9.9.210/32 is directly connected, Serial0/1/1
C       9.9.9.220/30 is directly connected, Serial0/2/0
L       9.9.9.221/32 is directly connected, Serial0/2/0
C       9.9.9.224/30 is directly connected, Serial0/2/1
L       9.9.9.225/32 is directly connected, Serial0/2/1

Router_Barbosa#
Router_Barbosa#
Router_Barbosa#
```

Router_Floridablanca

```
Router_Floridablanca>enable
Password:
Router_Floridablanca#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     9.9.9.129       YES NVRAM  up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/0   unassigned      YES unset  down                  down
Serial0/1/0            9.9.9.202       YES manual up                    up
Serial0/1/1            9.9.9.214       YES manual up                    up
Serial0/2/0            9.9.9.222       YES manual up                    up
Serial0/2/1            9.9.9.229       YES manual up                    up
Serial0/3/0            unassigned      YES unset  down                  down
Serial0/3/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Router_Floridablanca#
Router_Floridablanca#
```

```
Router_Floridablanca#
Router_Floridablanca#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     9.0.0.0/8 is variably subnetted, 15 subnets, 4 masks
S       9.9.9.0/28 [1/0] via 9.9.9.201
S       9.9.9.32/28 [1/0] via 9.9.9.201
S       9.9.9.64/28 [1/0] via 9.9.9.213
S       9.9.9.96/28 [1/0] via 9.9.9.221
C       9.9.9.128/28 is directly connected, GigabitEthernet0/0
L       9.9.9.129/32 is directly connected, GigabitEthernet0/0
S       9.9.9.160/27 [1/0] via 9.9.9.230
C       9.9.9.200/30 is directly connected, Serial0/1/0
L       9.9.9.202/32 is directly connected, Serial0/1/0
C       9.9.9.212/30 is directly connected, Serial0/1/1
L       9.9.9.214/32 is directly connected, Serial0/1/1
C       9.9.9.220/30 is directly connected, Serial0/2/0
L       9.9.9.222/32 is directly connected, Serial0/2/0
C       9.9.9.228/30 is directly connected, Serial0/2/1
L       9.9.9.229/32 is directly connected, Serial0/2/1

Router_Floridablanca#
```

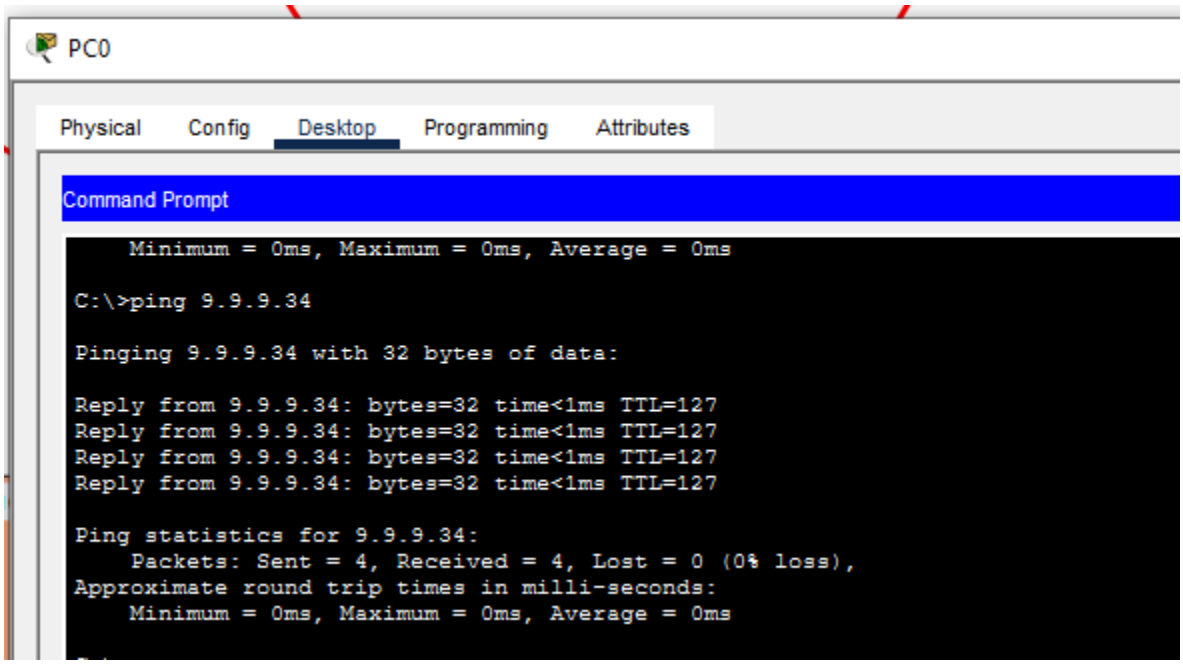
Router_Piedecuesta

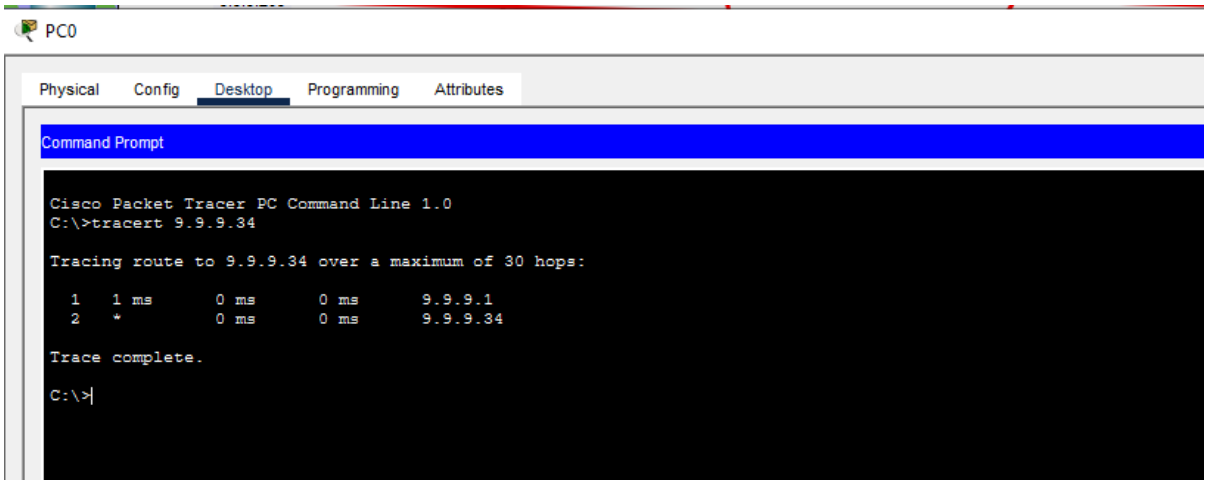
```
Router_Piedecuesta>en
Password:
Router_Piedecuesta#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     9.9.9.161       YES NVRAM  up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
FastEthernet0/0/0      unassigned      YES unset  down                  down
FastEthernet0/0/1      unassigned      YES unset  down                  down
FastEthernet0/0/2      unassigned      YES unset  down                  down
FastEthernet0/0/3      unassigned      YES unset  down                  down
Serial0/1/0            9.9.9.206       YES manual up                    up
Serial0/1/1            9.9.9.218       YES manual up                    up
Serial0/2/0            9.9.9.226       YES manual up                    up
Serial0/2/1            9.9.9.230       YES manual up                    up
Serial0/3/0            unassigned      YES unset  down                  down
Serial0/3/1            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Router_Piedecuesta#
```

```
Vlan1                  unassigned      YES unset  administratively down down
Router_Piedecuesta#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     9.0.0.0/8 is variably subnetted, 15 subnets, 4 masks
S       9.9.9.0/28 [1/0] via 9.9.9.205
S       9.9.9.32/28 [1/0] via 9.9.9.205
S       9.9.9.64/28 [1/0] via 9.9.9.217
S       9.9.9.96/28 [1/0] via 9.9.9.225
S       9.9.9.128/28 [1/0] via 9.9.9.229
C       9.9.9.160/27 is directly connected, GigabitEthernet0/0
L       9.9.9.161/32 is directly connected, GigabitEthernet0/0
C       9.9.9.204/30 is directly connected, Serial0/1/0
L       9.9.9.206/32 is directly connected, Serial0/1/0
C       9.9.9.216/30 is directly connected, Serial0/1/1
L       9.9.9.218/32 is directly connected, Serial0/1/1
C       9.9.9.224/30 is directly connected, Serial0/2/0
L       9.9.9.226/32 is directly connected, Serial0/2/0
C       9.9.9.228/30 is directly connected, Serial0/2/1
L       9.9.9.230/32 is directly connected, Serial0/2/1

Router_Piedecuesta#
Router_Piedecuesta#
```

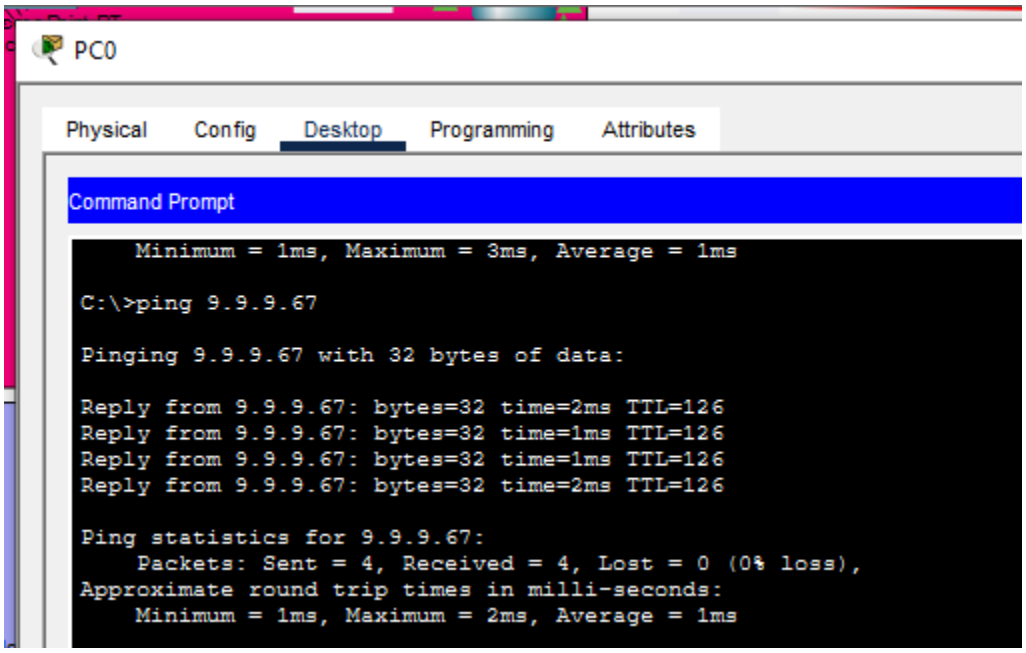
- Pruebas de conectividad (ping, tracert) desde una PC red 1 a las PC de las otras redes.

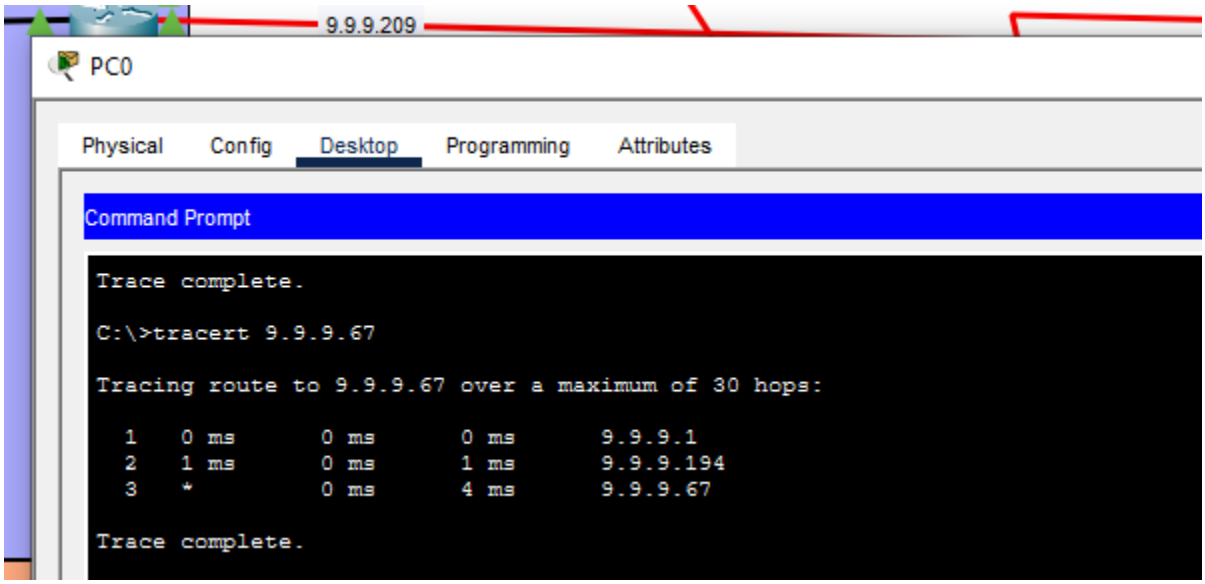
Desde PC0 (Red 1) a Printer3 (Red2)



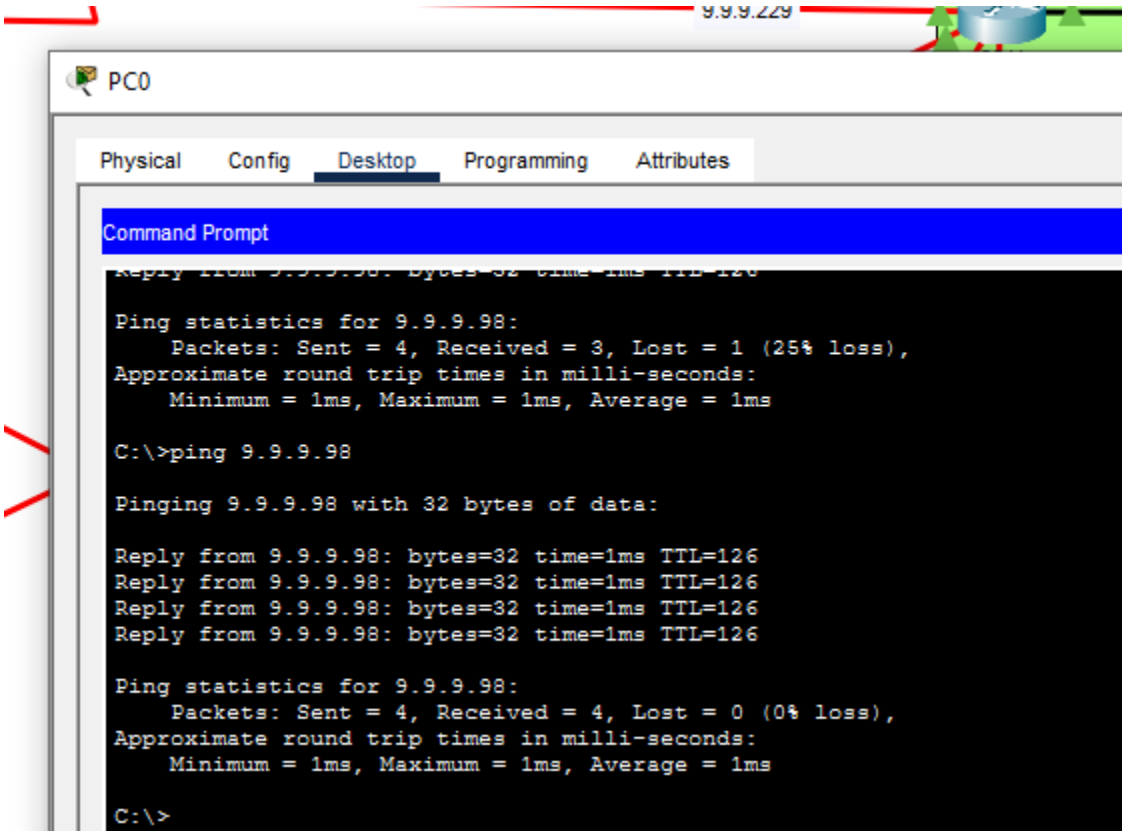


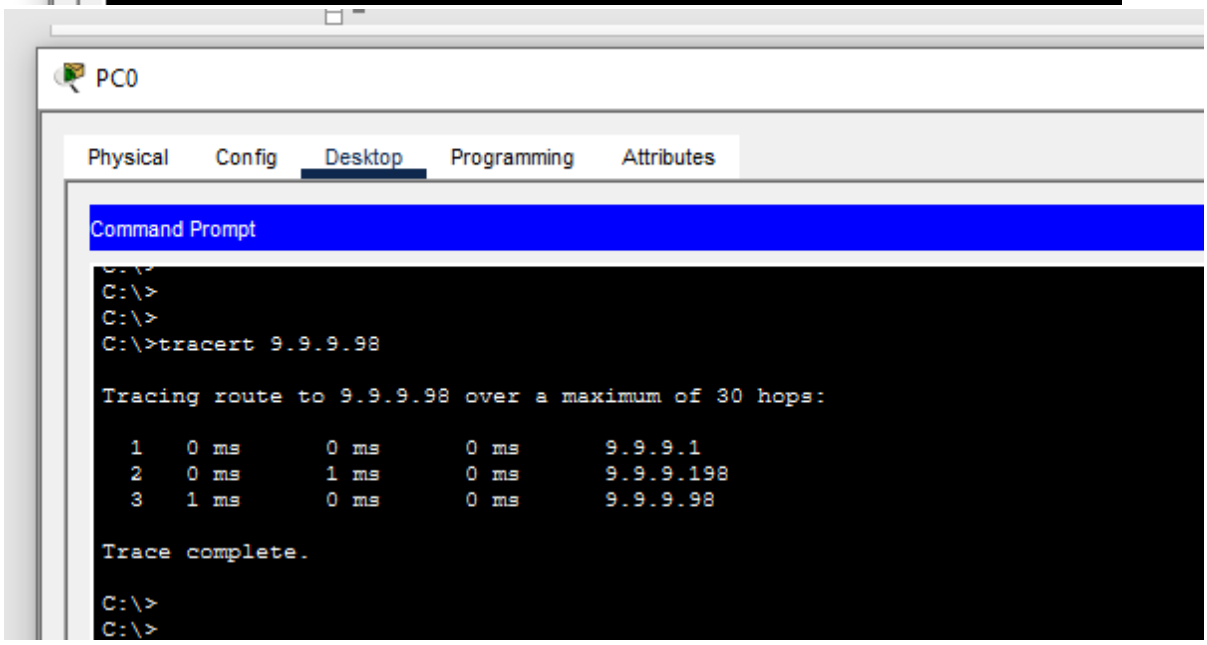
Desde PC0 (Red1) a Printer1 (Red 3)



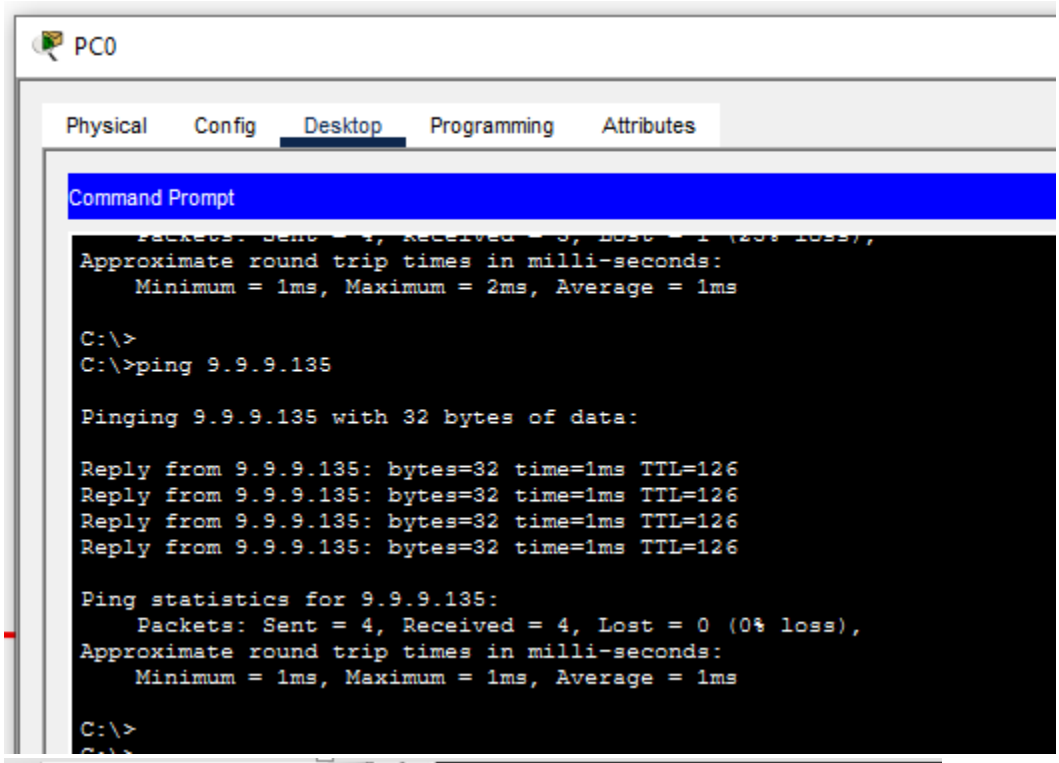


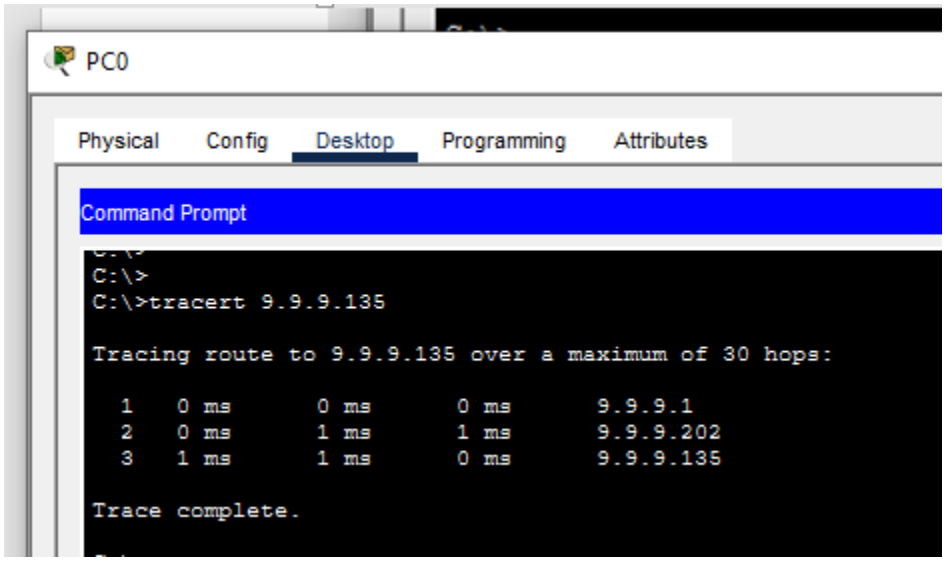
Desde PC0(Red1) a Laptop1(Red4)



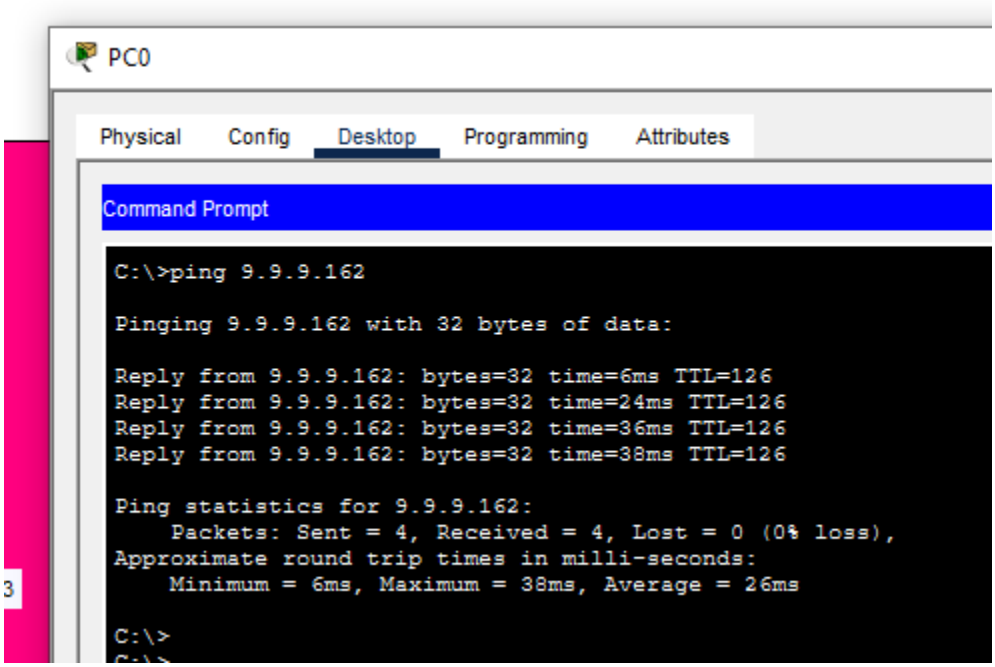


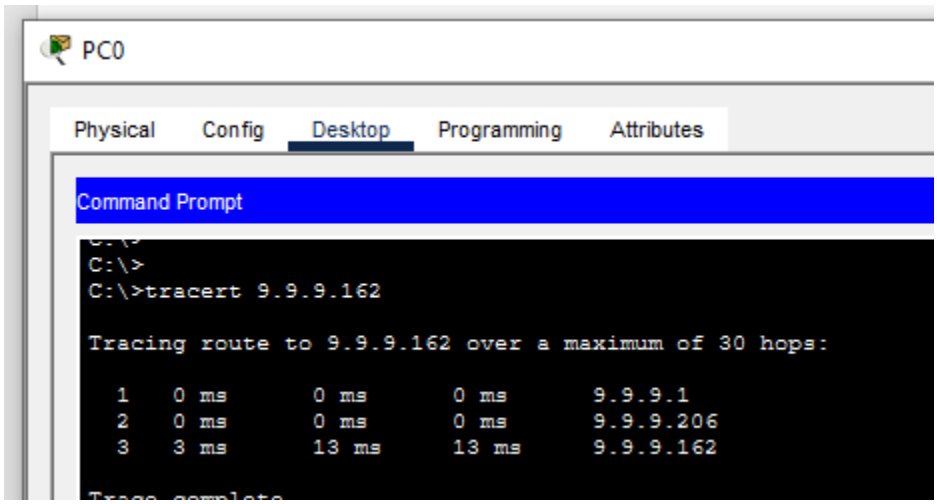
Desde PC0 (Red1) a PC6 (Red 5)





Desde PC0 (Red1) a Smartphone0 (Red6)





- Conexión FTP exitosa desde un router

Desde Router_Bucaramanga a FTP, copiando configuración de inicio

Router_Bucaramanga

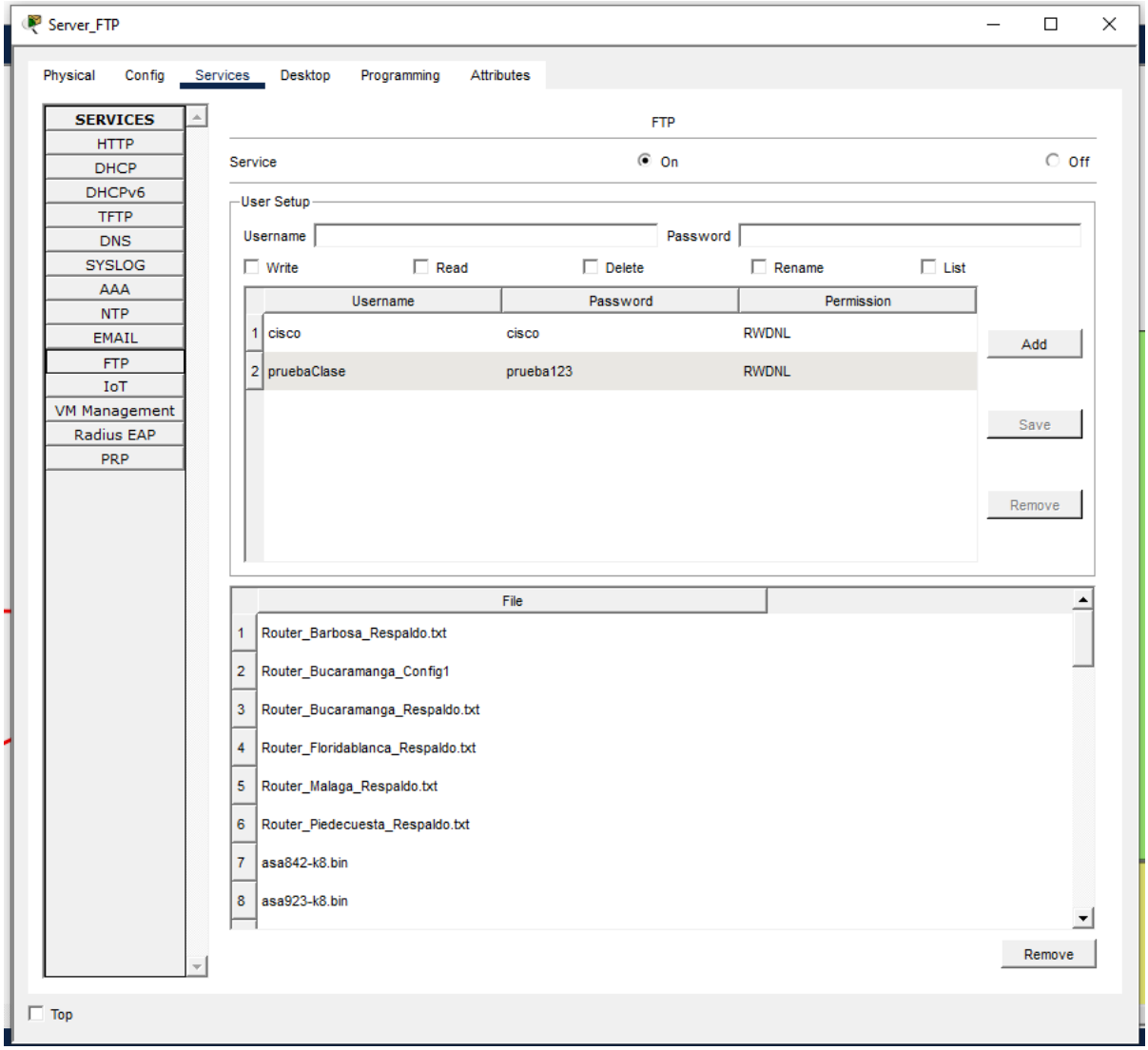
Physical Config CLI Attributes

IOS Command Line Interface

```
User Access Verification

Password:
% Password:  timeout expired!



Press RETURN to get started!


User Access Verification

Password:

Router_Bucaramanga>
Router_Bucaramanga>en
Password:
Router_Bucaramanga#copy running-config ftp
Address or name of remote host []? 9.9.9.130
Destination filename [Router_Bucaramanga-confg]? Router_Bucaramanga_Config1

Writing running-config...
[OK - 2055 bytes]

2055 bytes copied in 0.037 secs (55000 bytes/sec)
Router_Bucaramanga#
```

- Respaldo de routers al FTP

Guardando todas las configuraciones como Router_Respaldo.txt


Server_FTP
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
FTP
Service
On
Off
User Setup
Username
Password
Write
Read
Delete
Rename
List
Username
Password
Permission
1 cisco cisco RWDNL
2 pruebaClase prueba123 RWDNL
Add
Save
Remove
File
1 Router_Barbosa_Respaldo.txt
2 Router_Bucaramanga_Config1
3 Router_Bucaramanga_Respaldo.txt
4 Router_Floridablanca_Respaldo.txt
5 Router_Malaga_Respaldo.txt
6 Router_Piedecuesta_Respaldo.txt
7 asa842-k8.bin
8 asa923-k8.bin
Remove
Top